

Exponents and Radicals

$$a^n = \underbrace{a \cdot a \cdot \dots \cdot a}_{n \text{ copies}}$$

Similar to how:

$$n \cdot a = \underbrace{a + a + \dots + a}_{n \text{ copies}}$$

e.g. $3^1 = \underline{3}$

$$3^2 = \underline{3 \cdot 3} = 9$$

$$3^3 = \underline{3 \cdot 3 \cdot 3} = 27$$

$$3^0 = \underline{1}$$

$$3^{-1} = \underline{\frac{1}{3^1}} = \frac{1}{3}$$

$$3^{-2} = \frac{1}{3^2} = \frac{1}{9}$$

Rules of Exponents:

$$1) a^n \cdot a^m = a^{n+m}$$

$$2) \frac{a^n}{a^m} = a^{n-m}$$

$$3) (a^n)^m = a^{n \cdot m}$$

$$4) (a \cdot b)^n = a^n \cdot b^n$$

$$5) \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

e.g. Expand & simplify:

$$\left(\frac{2r^2}{s}\right)^3 \cdot \left(\frac{rs}{t}\right)^2$$

Radicals

If a is a real number, $\sqrt[n]{a}$ is a number so that $(\sqrt[n]{a})^n = a$.

$\sqrt[n]{a}$ may not always exist - e.g.
 $\sqrt[2]{-4}$.

In cases where there is more than one choice of value for $\sqrt[n]{a}$, there will be a pair of possible values, one positive and one negative - in this case we choose the positive value.
e.g. $\sqrt[2]{4} = \sqrt{4}$

If n is even, $\sqrt[n]{a}$ is defined only when $a \geq 0$ and is chosen to be nonnegative.

If n is odd, $\sqrt[n]{a}$ is defined for all real numbers, a , and has the same sign as a (unique value).

we may interpret radicals as fraction exponents, i.e. $\sqrt[n]{a} = a^{1/n}$.

$$\text{e.g. } \sqrt[3]{x} = x^{1/3}$$

$$\sqrt[2]{\frac{1}{3}} = \left(\frac{1}{3}\right)^{1/2} = 3^{-1/2}$$

$$\sqrt[5]{2^2} = \left(2^2\right)^{1/5} = 2^{2/5}$$

Using the rules of exponents we get similar rules for radicals:

$$\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\sqrt[n]{\sqrt[m]{a}} = \sqrt[n \cdot m]{a}$$

With radical problems we are typically trying to minimize the number of radicals present, reduce the amount of terms/shrink the expression inside the radicals present, or remove certain radicals.

e.g. $\sqrt[3]{16x^3y^8z^4}$ → simplify

$$= \sqrt[3]{16}$$

$$= \sqrt[3]{2^4 \cdot x^3 \cdot y^8 \cdot z^4}$$

$$= 2 \sqrt[3]{2 \cdot x^3 \cdot y^8 \cdot z^4}$$

$$= 2x \sqrt[3]{2 \cdot y^8 \cdot z^4}$$

$$= 2xy^2 \sqrt[3]{2 \cdot y^2 \cdot z^4}$$

$$= 2xy^2z \sqrt[3]{2 \cdot y^2 \cdot z}$$

$$= 2xy^2z \sqrt[3]{2y^2z}$$

$$(16x^3y^8z^4)^{1/3}$$

$$= 16^{1/3} x^1 y^{8/3} z^{4/3}$$

$$= 16^{1/3} (x^1) y^{2\frac{2}{3}} z^{1\frac{1}{3}}$$

$$(y^2)^{1/3} y^{2/3}$$

$$= x^1 y^2 z^1 16^{1/3} y^{2/3} z^{1/3}$$

$$2^{4/3} (2^1)^{1/3}$$

$$\sqrt[3]{y^2 \cdot z}$$

e.g. $\sqrt[5]{\frac{x}{y^2}}$ $\rightarrow$ rationalize the denominator

$$= \frac{\sqrt[5]{x}}{\sqrt[5]{y^2}} = \frac{\sqrt[5]{x} \cdot \sqrt[5]{y^3}}{\sqrt[5]{y^2} \cdot \sqrt[5]{y^3}}$$

$$n=5$$

$$b=y$$

$$k=2$$

$$\frac{\sqrt[5]{x} \cdot \sqrt[5]{y^3}}{y} = \frac{\sqrt[5]{x \cdot y^3}}{y}$$

e.g. $\left(\frac{x^6}{9y^{-4}} \right)^{-1/2} \rightarrow$ simplify

$$\frac{(x^6)^{-1/2}}{9^{-1/2}(y^{-4})^{-1/2}}$$

$$\sqrt{9} = 3$$

$$9^{1/2} = 3$$

$$9^{-1/2} = \frac{1}{3}$$

$$\frac{3(x^6)^{-1/2}}{(y^{-4})^{-1/2}} = \frac{3|x|^{-3}}{|y|^2} = \frac{3|y|^{-2}}{|x|^3}$$